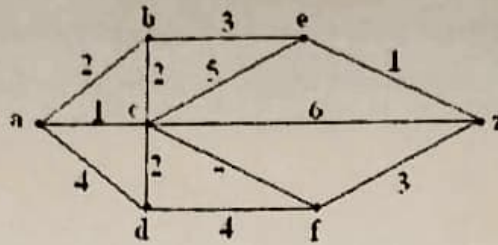
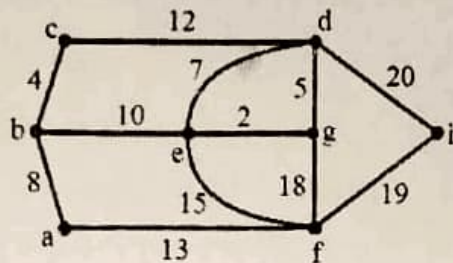


- (c) The Lucas numbers satisfies the recurrence relation $L_n = L_{n-1} + L_{n-2}$, with $L_0 = 2$ and $L_1 = 1$. Find an explicit formula for the Lucas numbers. (06)

- Q4. (a) Find the length of a shortest path between a and z in the given weighted graph, using Dijkstra's algorithm: (06)

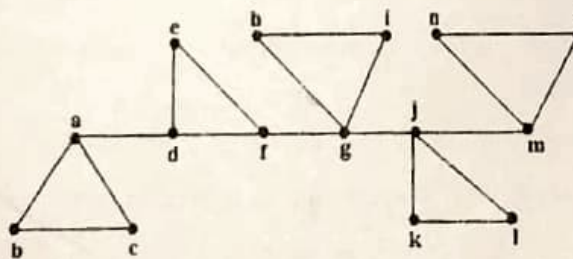


- (b) Determine a minimum spanning tree in the following weighted graph using Kruskal's algorithm: (06)



- (c) (i) Represent the compound proposition $(\sim p \wedge (q \leftrightarrow \sim p)) \vee (\sim q)$ using an ordered rooted tree. Also evaluate their prefix, postfix and infix notations. (04)

- (ii) Define Chromatic number. Find the chromatic number of the given graph: (02)



- Q5: (a) If G be the set of four special bilinear functions f_1, f_2, f_3, f_4 on the set of complex numbers defined by $f_1(z) = -z$, $f_2(z) = 1/z$, $f_3(z) = z$ and $f_4(z) = -1/z$. Then prove that G forms a finite abelian group of order 4 with respect to composition known as composite of two functions. (06)

- (b) Show that the set of numbers of the form $a + b\sqrt{2}$, with a and b as rational numbers, is a field. (06)

- (c) (i) Express the following into disjunctive normal form: (03)

$$f(x, y, z) = (x + \bar{y})(y + \bar{z})(z + \bar{x})(\bar{x} + \bar{y})$$

- (ii) Simplify the expression and draw a logic circuit and switching circuit of the Boolean function: (03)

$$f(r, s, t) = r.t + [s.(s' + t). \{r' + (s.t)\}]$$